STAFF REVIEW · H446 288 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Encryption.

2. Acts as an intermediary for requests/responses.

3. Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

4. Q changes to 1 on the next rising edge, not simply when D changes.

I Do — source worked steps

1. Predict first — Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

2. Show method — Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

We Do — guided CHECK

1. Predict first
Alice encrypts to Bob's public key without needing to learn Bob's private key.
Show method
Bob retains the private key and decrypts. The public key must be authenticated to prevent substitution.

You Do — source mark schemes

1. Symmetric encrypt/decrypt use shared secret [2]; asymmetric uses public/private pair [2], encrypt to recipient's public key [1], private key decrypts [1].

2. Server identity authenticated via certificate [2]; authenticated key agreement establishes session secret [2]; symmetric session encryption protects bulk traffic efficiently [2].

3. Fast for bulk data [1]; each shared secret needs secure exchange [2]; managing many unique shared keys is difficult [1]; asymmetric authenticated exchange/hybrid design improves distribution [2].

Exit Ticket — source answer

1. False. The paired private key is secret; the public key needs authentic distribution.